Tony Carricarte

Data Scientist - Generative AI

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Professional Experience

2026–present **Data Scientist — Generative AI**, *OncoBrain*, USA
- Designed and deployed a RAG system for clinical decision support in oncology, delivering evidence-based, contextually grounded treatment recommendations.
- Engineered a retrieval evaluation framework benchmarking full-text, vector, hybrid, and reranked-hybrid search strategies across quality and latency metrics, enabling data-driven selection of retrieval pipelines.
- Built and maintained ETL pipelines ingesting multiple oncology guideline sources, resolving integration and runtime failures to improve data reliability and metadata quality.

2025–2026 **Neuromatch Impact Scholar Program**, Germany
- Developed representational learning framework for medical image segmentation using a transformer-based architecture (SAM).

2023–2025 **Teaching Assistant**, *Free University of Berlin*, Germany
- Taught "Programming Foundations for Psychological Science" to undergraduate psychology students.
- Designed course materials covering object-oriented programming and core data-science libraries in Python.

2020–2025 **Research Scientist**, *Free University of Berlin*, Germany
- Conducted deep-learning research on human object vision using CNNs, time-series EEG, and ultra-high-resolution neuroimaging, resulting in two peer-reviewed publications.
- Developed scalable, HPC-based processing pipelines for large MRI datasets using Python and Linux clusters.
- Investigated tokenization approaches in LLMs to improve transfer learning for sentiment classification.
- Led international collaborations across academic institutions, communicating complex modeling results to interdisciplinary teams.
- Supervised one intern on modeling and data-analysis techniques.

Education

2020–2025 **PhD in Computational Neuroscience**, *Free University of Berlin*, Germany

2017–2019 **MSc in Neuroscience (Joint)**, *Universities of Göttingen and Bordeaux*, Germany / France

2010–2015 **BSc in Biology**, *University of Havana*, Cuba, *Summa cum laude*

Skills

AI Engineering: Python, PyTorch, Pandas, NumPy, Scikit-learn, LanceDB, LangChain, Git, FastAPI, SQL, Claude Code

AI Modeling: LLMs, RAG, Vision Transformers, Representational learning, Feature extraction, Classification, Regression, A/B testing, Fine-tuning, XGBoost, End-to-end training and validation, CI/CD pipelines

Languages: Spanish (Native), English (Fluent), German (Fluent)

Awards, Grants and Scholarships

2024 Einstein Center for Neurosciences Graduation Scholarship — 10,800 €

2019–2022 Einstein Center for Neurosciences PhD Fellowship — 63,000 €

2017–2019 Erasmus Mundus Joint Master Degree Scholarship — 49,000 €

Publications

Peer-reviewed **Carricarte T** et al. (2026). Layer-specific spatiotemporal dynamics of feedforward and feedback in human visual object perception. *eLife*.

Peer-reviewed **Carricarte T** et al. (2024). Laminar dissociation of feedforward and feedback in high-level ventral visual cortex during imagery and perception. *iScience*.